ELECTRICAL—POWER WINDOW—POWER WINDOW SHORT DROP GLASS FEATURE INOPERATIVE OR ERRATIC OPERATION	Article No. 02-5-7

FORD: 2002 THUNDERBIRD

ISSUE

Some vehicles may exhibit calibration loss or erratic operation of the Power Window Short Drop Glass function. This may be caused by a discharged or weak battery, presence of door glass mechanical interference, or by the software program in the Front Electronics Module (FEM) and Rear Electronics Module (REM).

ACTION

Test the function and operation of the Power Window Short Drop Glass feature. Use the following Service Procedure to determine the best suited repair action.

SERVICE PROCEDURE

System Features:

In order to improve door efforts and sealing, the 2002 Thunderbird is equipped a Short Drop Glass feature. This feature lowers either the left or right door window approximately 10 mm when the door is opened. The door window returns to its full up position when the door is closed. If the optional removable hard top is not installed on the vehicle, this feature will also lower the windows in both doors when the convertible top switch is operated. Both door windows will return to their full up position if the transmission is shifted out of "P" (Park).

Proper operation of the Short Drop Glass requires that the windows be calibrated. This must be done to both door windows independently since each is commanded by a separate module. Even though the Short Drop Glass feature is calibrated from the factory, it is possible for the windows to lose calibration during operation in service.

NOTE

BEFORE ATTEMPTING TO DIAGNOSE A PERCEIVED CONCERN WITH THE SHORT DROP GLASS FUNCTION, CHECK THE BASIC OPERATION OF BOTH POWER WINDOWS USING THE SUGGESTIONS LISTED BELOW.

CORRECT THE OBSERVED CONDITION AS NEEDED TO RESTORE CORRECT POWER WINDOW OPERATION. DO NOT CONTINUE WITH THIS ARTICLE IF A BASIC POWER WINDOW FUNCTIONALITY CONCERN EXISTS.

- Check the glass for tilting or other potential adjustment issues while operating the power windows. If the glass requires adjustment refer to Workshop Manual, Section 501-11, for details.
- Inspect window seals and runs that may be mispositioned causing power window operation concern. Adjust, clean and lubricate as needed to restore proper operation.
- Inspect the roof seals for correct positioning. The system may identify out of position or "drooping" roof seals as a foreign object and lose calibration. Service as needed to repair. Refer to Workshop Manual, Section 501-18, for details.
- Inspect the door window regulator for attachment issues, cables binding or misrouted and other potential operational issues. Service as needed to restore operation. Refer to Workshop Manual, Section 501-11, for details.

1. Verify the customer concern and identify the mode in which the Short Drop Glass feature is defaulted. If the system is presently operating, do not calibrate the system. Calibrating an operational system is not required and will not improve system performance.

There are two (2) states in which the system may have defaulted.

State 1: Short Drop Glass feature will lower the window, but will not raise it again when the door is closed. Closing and opening the door again may cause the window to drop further, but the window will not raise to full up position.

Calibration Procedure for State 1: Press the Power Window Switch "up" on the effected door to raise the window completely and continue to hold the switch in the "up" position for 2 seconds.

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State 2: Short Drop Glass feature will not lower the window when the door is opened. When the window is in the full "up" position, the door will be difficult to open and close because the roof & A-pillar seals are in close contact with the window.

Calibration Procedure For State 2: Press the Power Window switch "down" on the effected door to lower the window completely and continue to hold the switch in the "down" position for 2 seconds. Immediately after releasing the window "down" switch, press the "up" switch to raise the window completely and continue to hold the switch in the "up" position for 2 seconds.

NOTE
DURING SHORT DROP GLASS CALIBRATION, THE AFFECTED DOOR MUST BE CLOSED AND THE TOP MUST BE INSTALLED (IF USING THE HARD TOP) OR BE IN THE UP POSITION (IF USING CONVERTIBLE TOP).

NOTE
THE LEFT AND RIGHT POWER WINDOW MOTORS ARE CONTROLLED BY 2 (TWO) SEPARATE MODULES AND THEREFORE MUST BE CALIBRATED INDEPENDENTLY. IF BOTH WINDOWS HAVE LOST THEIR CALIBRATION SIMULTANEOUSLY, A DEGRADATION OR LOSS OF BATTERY POWER SHOULD BE SUSPECTED. FEM CONTROLS THE LEFT WINDOW, REM CONTROLS THE RIGHT WINDOW.

2. If this is the first and only occurrence the customer has experienced with a loss of Short Drop Glass function, simply calibrate the system based on the symptoms listed above and release the vehicle back to the customer. Please refer the customer to the Power Windows section of the Owner's Guide for future calibration requirements. It is normal for the system to lose calibration if one or more of the following events took place:
   - Degradation or loss of battery power
   - Door Glass was closed on a foreign object
   - Door Glass was frozen with snow and ice. Power Window Switch was toggled up/down multiple times in an effort to free window.
3. If the Short Drop Glass function has been lost on several occasions, test for a possible mechanical concern with the effected power window mechanism using the following procedure:
   a. Bring vehicle indoors and allow it to warm to approximately 20° C (68° F).
   b. Use a stop watch to measure how long it takes the window to run through its down and up movements.
   c. Battery fully charged, engine running and the doors closed, it should take no longer than 4 seconds for the window to roll down and no longer than 5 seconds for the window to roll up.
4. If power window performance does not meet this specification, check for possible window adjustment or mechanical issues discussed in the beginning notes of this article. If power window performance does meet this specification, continue to Step 5.
5. If it is determined that no battery power and mechanical issues are present, and the condition is a source of dissatisfaction with the customer, replace the FEM & REM which have upgraded software. The new modules may reduce the frequency in which window calibration will be required, however it should be communicated to the customer that the Short Drop Glass feature may require periodic calibration. Use the following service procedure to install a new FEM and REM:
   a. Use NGS Service Card version 10.1 or higher only.
   b. Select "Programmable Module Installation".
   c. Choose "FEM" or "GEM".
   d. Select "Retrieve Data - Old ECU".
   e. Remove the original module and install new FEM, part number 1W6Z-14B205-AA. Refer to Workshop Manual, Section 419-10, for details.
   f. Select "Programmable Module Installation".
   g. Choose "FEM" or "GEM".
   h. Select "Restore Configuration - New ECU".
   i. Select "Customer Preference Items".
   j. Select "Module Option Content".
   k. Select "2002 Thunderbird".
   l. Select the following "Short Drop Down Error", "Short Drop Error", "Hall Short Error".

m. Set selections to "Enable" and select "Done".

n. Replace the REM, part number 1W6Z-13C788-AB, repeating steps "a" through "m". Refer to Workshop Manual, Section 419-10, for details. Simply select "REM" in steps "c" and "g".

o. Recalibrate the Short Drop Glass on both windows using Step 1 - State 2.

PART NUMBER	PART NAME
1W6Z-14B205-AA	Front Electronics Module (FEM)
1W6Z-13C788-AB	Rear Electronics Module (REM)

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: Eligible Under The Provisions Of Bumper To Bumper Warranty Coverage

OPERATION	DESCRIPTION	TIME
020507A	Verify Concern And Recalibrate Power Windows (Includes Time To Perform Procedure For State One And Two On Both Windows)	0.1 Hr.
020507B	Verify Concern And Recalibrate Power Windows (Includes Time To Perform Procedure For State One And Two On Both Windows) Then Perform Test And Replace F.E.M And R.E.M.	1.4 Hrs.

DEALER CODING

BASIC PART NO.	CONDITION CODE
14B205	42

OASIS CODES: 102000